

# University of Natural Resources and Life Sciences, Vienna (BOKU)

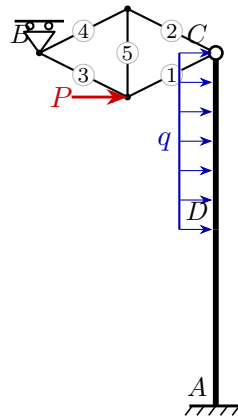
## VU Mechanik – LAWI 100515 (PI)

Exam **Group A**

Date: 26.06.2026

### Given system

The planar system consists of bending beams and a truss connected by hinges.



**Given quantities:**  $a = 3 \text{ m}$   $\ell = 1,5 \text{ m}$   $h = 1,5 \text{ m}$   $q = 5 \text{ kN/m}$   $P = 15 \text{ kN}$

### Tasks

1. Determine the degree of static determinacy.
2. Compute all support reactions and hinge forces.
3. Determine all truss member forces (tension/compression).
4. Draw the section forces  $N$ ,  $V$ ,  $M$  of the bending beams.